

Interaction of high-energy particles: Electrons/positrons (Energy loss processes) (Lecture 4)

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Outline of this lecture

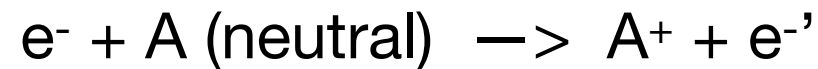
- Interaction of electrons/positrons with matter
 - Ionization/excitation, Coulomb collision
 - Bremsstrahlung
 - Electron-positron annihilation
- Interaction with radiation field
 - Inverse Compton loss
- Interaction with magnetic field
 - Synchrotron loss
 - Deflection/scattering (will discussed later)

Interaction of high-energy particles: electrons/positrons

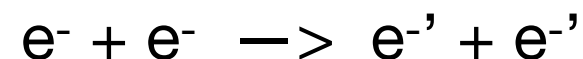
Interaction with matter

Four major interaction processes with matter

- Ionization/excitation of atoms in neutral medium:



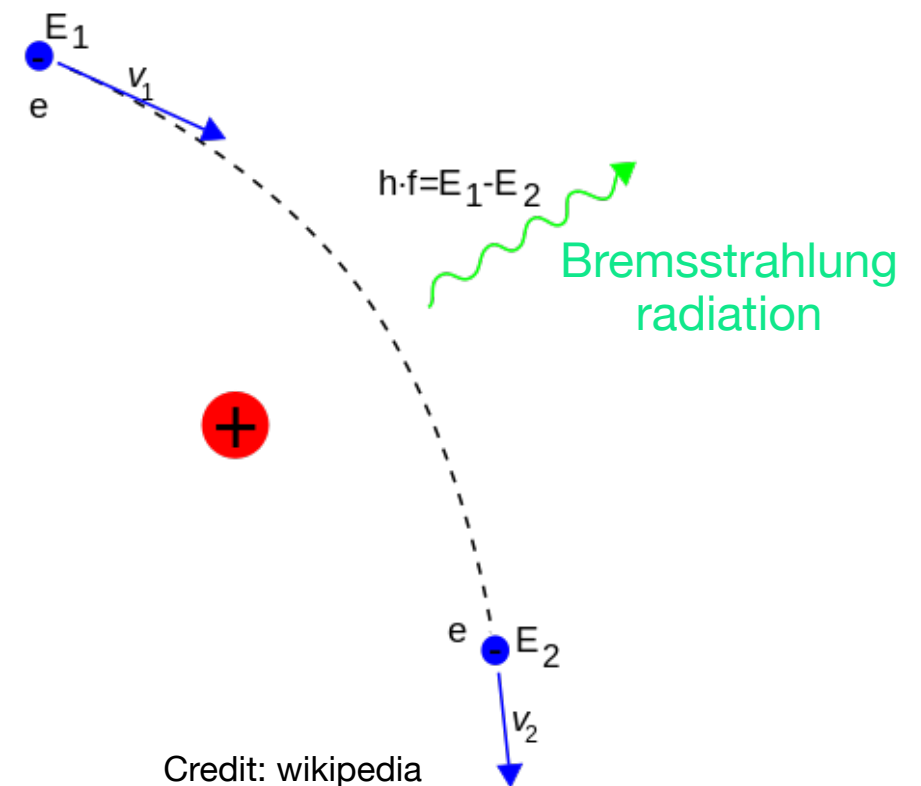
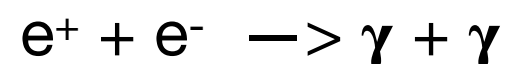
- Coulomb scattering off the thermal electrons in ionized medium (plasma):



- Bremsstrahlung (de-acceleration)

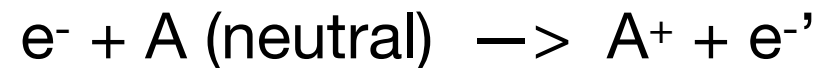
due to medium electrons/nuclei:

- Annihilation of electron and positron:



Interaction of high-energy electrons/positrons with matter

Ionization/excitation in neutral medium



Energy loss rate due to ionization: dE/dt

Ionization losses in the neutral hydrogen and helium are given by the Bethe-Bloch formula

$$\left(\frac{dE}{dt}\right)_I = -2\pi r_e^2 c m_e c^2 \frac{1}{\beta} \sum_{s=\text{H,He}} Z_s n_s \left\{ \ln \left[\frac{(\gamma - 1)\beta^2 E^2}{2I_s^2} \right] + \frac{1}{8} \right\}$$

where Z_s is the nucleus charge, n_s is the gas number density,

I_s is the ionization potential $I_{\text{H}} = 13.6 \text{ eV}$, $I_{\text{He}} = 24.6 \text{ eV}$

E is the total electron energy,

γ and $\beta = v/c$ are the electron Lorentz factor and speed, respectively.

$r_e = 2.8 \times 10^{-15} \text{ m}$ is the classical electron radius.

Interaction of high-energy electrons/positrons with matter

Coulomb scattering/collisions in ionized medium (plasma)

$$e^- + e^- \rightarrow e'^- + e'^-$$

- Scattering off thermal electrons present in the medium.

Energy loss rate due to coulomb collisions: dE/dt

The Coulomb energy losses in the fully ionized medium in the cold plasma limit are described by

$$\left(\frac{dE}{dt}\right)_{\text{Coul}} = -2\pi r_e^2 c m_e c^2 n_e \frac{1}{\beta} \left[\ln \left(\frac{E m_e c^2}{4\pi r_e \hbar^2 c^2 n_e} \right) - \frac{3}{4} \right],$$

where n_e is the electron number density in the plasma

$r_e = 2.8 \times 10^{-15}$ m is the classical electron radius.

E is the total electron energy,

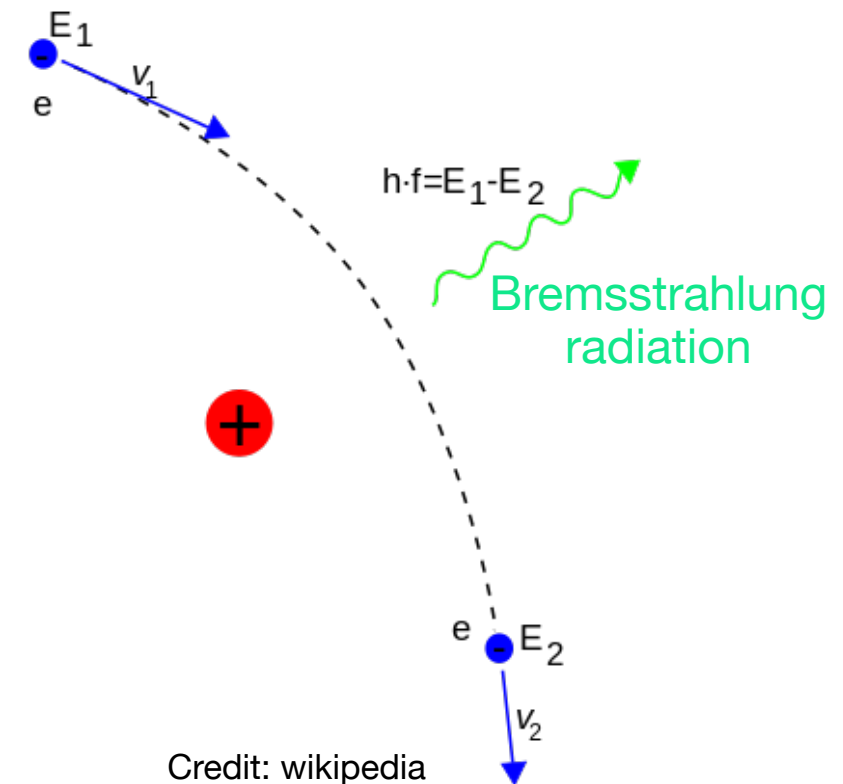
$\hbar \equiv h/2\pi$ is the Planck constant,

$\beta = v/c$

Interaction of high-energy electrons/positrons with matter

Bremsstrahlung (de-acceleration)

- 1930s — The ionization loss rate by Bethe-Bloch formula underestimates the energy loss rate for relativistic electrons.
- An additional energy loss happens due to the acceleration/deacceleration of the electron in the field of the nucleus.
- Sometimes referred to as *free-free* emission: Transition between unbound states of an electron in the field of the nucleus.
- Nikola Tesla (1880s), Bethe & Heitler (1934), Koch & Motz (1959), Blumenthal & Gould (1970).



Interaction of high-energy electrons/positrons with matter

Bremsstrahlung (de-acceleration)

Bremsstrahlung energy loss rate of an electron of total energy E and Lorentz factor γ in an ionized plasma:

$$\left(\frac{dE}{dt}\right)_{BI} = -4\alpha_f r_e^2 c m_e c^2 Z(Z + 1)nE \left[\ln(2\gamma) - \frac{1}{3} \right]$$

where Z and n are the charge number and number density of the ion in the plasma.

Interaction of high-energy electrons/positrons with matter

Bremsstrahlung (de-acceleration)

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where Z and n are the charge number and number density of the ion in the plasma.

Bremsstrahlung energy loss rate in neutral gas:

$$\left(\frac{dE}{dt}\right)_{B0} = \begin{cases} -4\alpha_f r_e^2 c m_e c^2 E \left[\ln(2\gamma) - \frac{1}{3} \right] \sum_{s=\text{H,He}} n_s Z_s (Z_s + 1), & \gamma \lesssim 100 \\ -cE \sum_{s=\text{H,He}} \frac{n_s M_s}{T_s}, & \gamma \gtrsim 800, \end{cases}$$

with a linear connection in between.

M_s is the atomic mass

T_s is the radiation length ($T_{\text{H}} \simeq 62.8 \text{ g cm}^{-2}$, $T_{\text{He}} \simeq 93.1 \text{ g cm}^{-2}$).

n_s and Z_s are the number density and charge number of an element s .

Electron-positron annihilation

$$e^+ + e^- \rightarrow \gamma + \gamma$$

- **Annihilation at rest:** Both photons have energies equal to 0.511 MeV (the rest-mass energy of an electron).

Electron-positron annihilation

$$e^+ + e^- \rightarrow \gamma + \gamma$$

- **Annihilation in flight:** There is a dispersion in photon energies.

E.g: if the positron is moving with a velocity v and Lorentz factor γ , and the electron is at rest, the ejected photons have energies given by

$$E = \frac{m_e c^2 (1 + \gamma)}{2} \left(1 \pm \frac{V}{c} \right) \quad \begin{array}{l} + \rightarrow \text{in the direction of the positron} \\ - \rightarrow \text{opposite} \end{array}$$

$$\text{where } V = \gamma v (1 + \gamma)$$

Electron-positron annihilation

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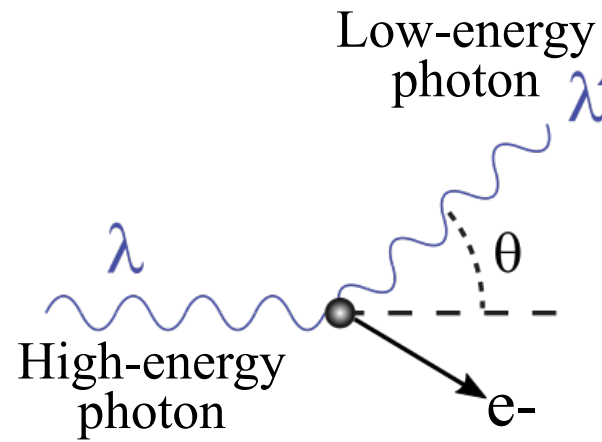
$$E = \frac{m_e c^2 (1 + \gamma)}{2} \left(1 \pm \frac{V}{c} \right) \quad \begin{array}{l} + \rightarrow \text{in the direction of the positron} \\ - \rightarrow \text{opposite} \end{array}$$

$$\text{where } V = \gamma v (1 + \gamma)$$

- The cross-section in extreme relativistic limit is: $\sigma = \frac{\pi r_e^2}{\gamma} [\ln 2\gamma - 1]$
- At thermal energies: $\sigma \approx \frac{\pi r_e^2}{(v/c)}$ where $r_e = 2.8 \times 10^{-15}$ m is the classical electron radius.
- Interaction time scale, $t_{\text{anni}} = 1/(n_e v \sigma)$ where n_e : ambient electron density.

Interaction of high-energy electrons/positrons with radiation field

Compton scattering



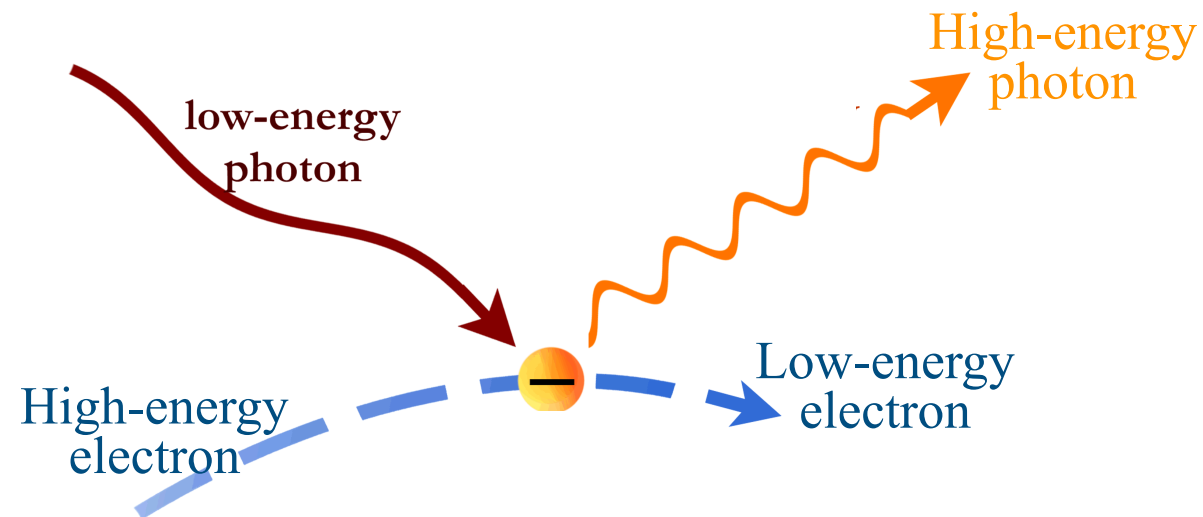
$$\lambda' - \lambda = \frac{h}{m_e c} (1 - \cos \theta),$$

- λ is the initial wavelength,
- λ' is the wavelength after scattering,
- h is the **Planck constant**,
- m_e is the **electron rest mass**,
- c is the **speed of light**, and
- θ is the scattering angle.

High-energy photon scatters off an electron: Photon losses energy.
Electron gains energy.

Interaction of high-energy electrons/positrons with radiation field

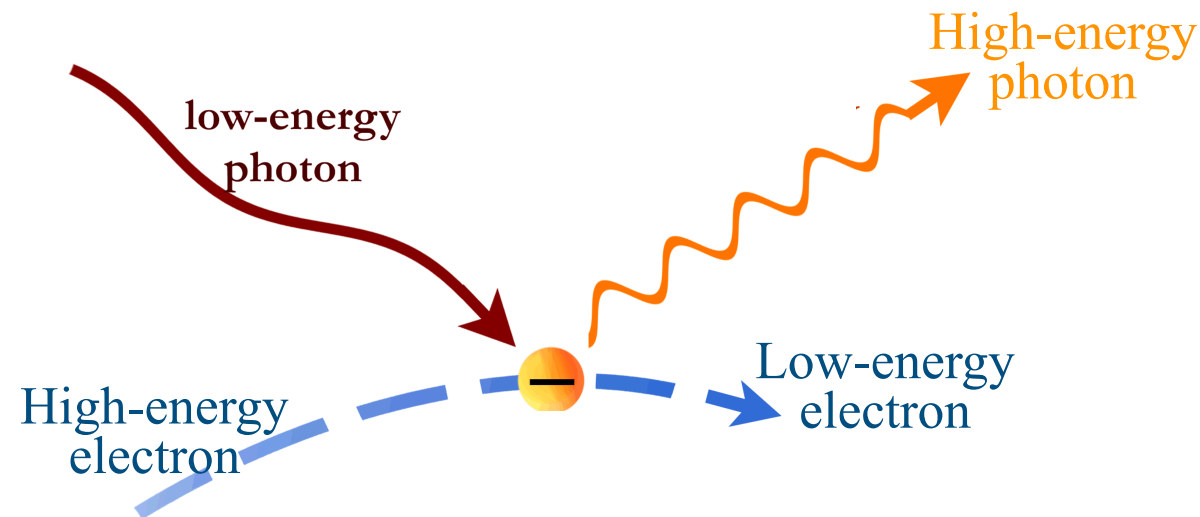
Inverse Compton process



- * High-energy electron scatters a low-energy photon: Photon gains energy.
Electron loses energy.
- * The process is opposite to the Compton process. So, the name “inverse Compton”.

Interaction of high-energy electrons/positrons with radiation field

Inverse Compton process



*Energy loss rate:
$$-\left(\frac{dE}{dt}\right)_{\text{IC}} = \frac{4}{3} \sigma_{\text{T}} c u_{\text{rad}} \left(\frac{v^2}{c^2}\right) \gamma^2$$

where $\sigma_{\text{T}} = 6.653 \times 10^{-29} \text{ m}^2$ is the Thompson crosssection

u_{rad} : Energy density of the target photons

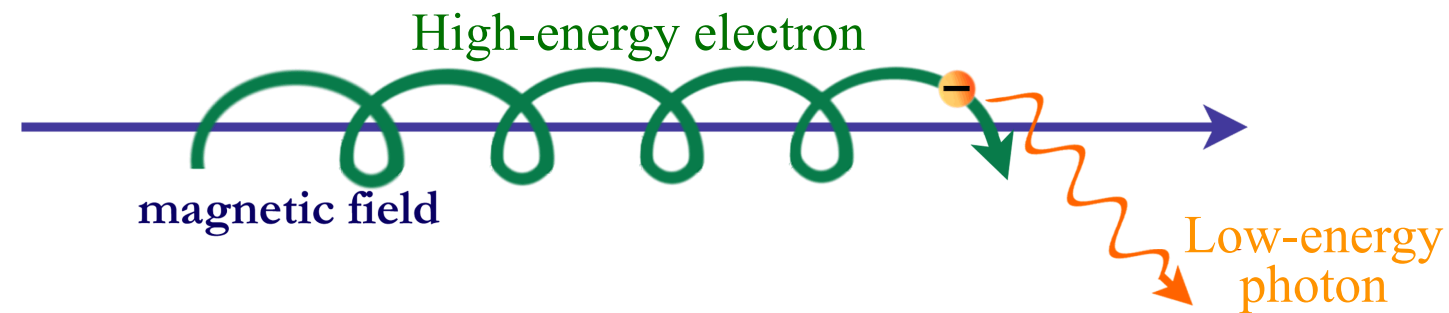
v : velocity of the incoming electron

γ : Lorentz factor of the incoming electron

*More accurate formula using Klein-Nishina crosssection is given in Strong & Moskalenko, 1998, ApJ, 509, 212 (Equations C13).

Interaction of high-energy electrons/positrons with magnetic field

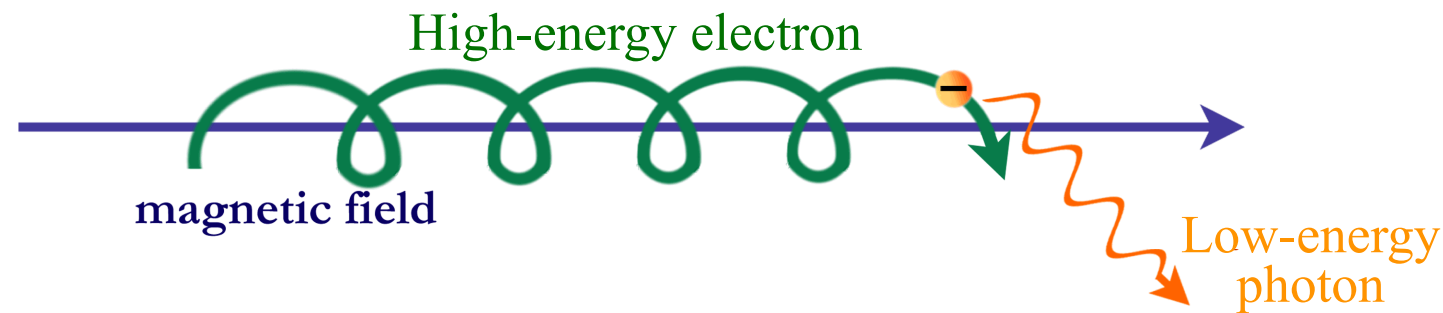
Synchrotron loss process



High-energy electron gyrating in a magnetic field emits radiation:
- Electron loses energy.

Interaction of high-energy electrons/positrons with magnetic field

Synchrotron loss process



High-energy electron gyrating in a magnetic field emits radiation:
- Electron loses energy.

$$\text{*Energy loss rate: } -\left(\frac{dE}{dt}\right)_{\text{sync}} = \frac{4}{3}\sigma_T c u_{\text{mag}} \left(\frac{v^2}{c^2}\right) \gamma^2$$

where $\sigma_T = 6.653 \times 10^{-29} \text{ m}^2$ is the Thompson crosssection

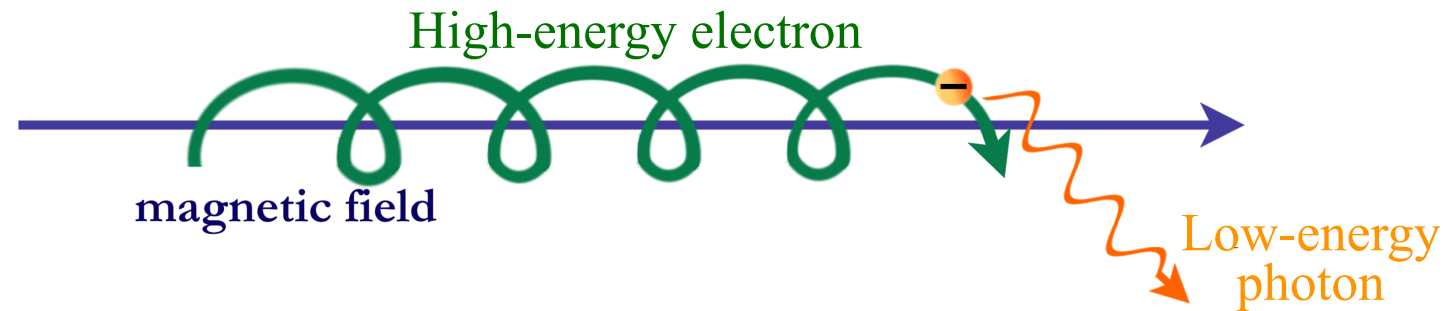
u_{mag} : Energy density of the magnetic field

v : velocity of the incoming electron

γ : Lorentz factor of the incoming electron

Interaction of high-energy electrons/positrons with magnetic field

Synchrotron loss process



High-energy electron gyrating in a magnetic field emits radiation:
- Electron loses energy.

$$\text{*Energy loss rate: } -\left(\frac{dE}{dt}\right)_{\text{sync}} = \frac{4}{3} \sigma_T c u_{\text{mag}} \left(\frac{v^2}{c^2}\right) \gamma^2$$

where $\sigma_T = 6.653 \times 10^{-29} \text{ m}^2$ is the Thompson crosssection

u_{mag} : Energy density of the magnetic field

v : velocity of the incoming electron

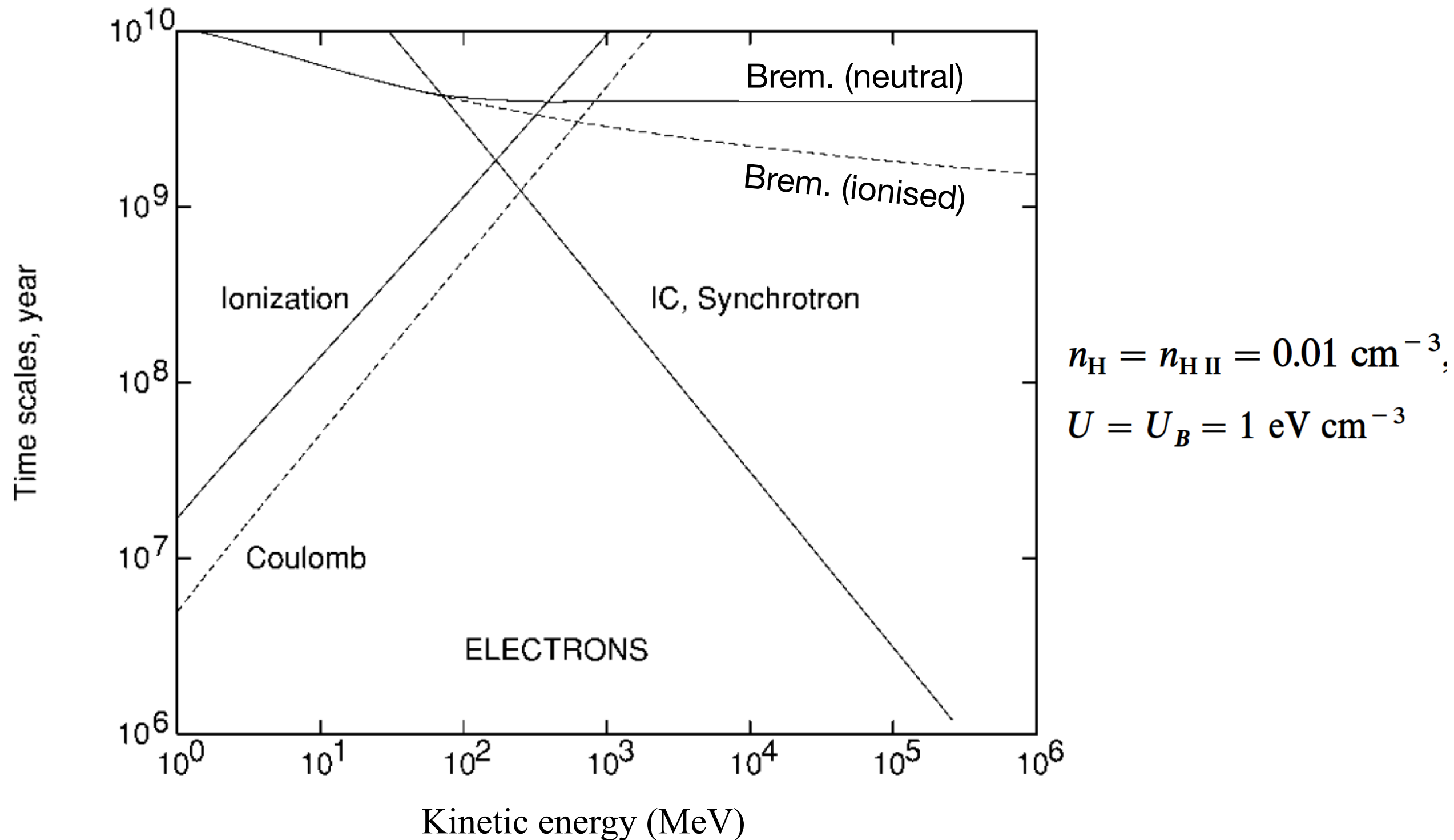
γ : Lorentz factor of the incoming electron

*Notice the similarity to the energy loss rate by inverse Compton process:

$$-\left(\frac{dE}{dt}\right)_{\text{IC}} = \frac{4}{3} \sigma_T c u_{\text{rad}} \left(\frac{v^2}{c^2}\right) \gamma^2$$

Interaction of high-energy particles: electrons/positrons

Energy loss timescale, $t_{\text{loss}} = E/(dE/dt)$ where E is the kinetic energy.



Next, we will learn about the acceleration mechanisms of high-energy particles.